

Spatial Competition with General Intensive-Margin Demand

Jonathan Vogel

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Abstract

Spatial competition models typically focus on extensive-margin spatial allocations of consumers across suppliers, often imposing unit demand. This paper develops a general reduction for incorporating intensive-margin demand in spatial environments. With common two-part tariffs, each tariff induces a single tariff index. When consumer-firm fixed match terms enter additively and independently of quantity, this tariff index is the only firm-level tariff object relevant for consumer assignment, and the resulting two-stage game is payoff-equivalent to an associated unit-demand spatial game with transformed firm types. We apply the general results to characterize a set of subgame-perfect equilibria in a heterogeneous-firm circular model with arbitrary intensive demand.

1 Introduction

Spatial competition models—whether in product-characteristic or geographic space—are organized around the extensive-margin spatial allocation of consumers across firms: the central object is to understand how spatial frictions divide consumers across competitors and shape firm-level outcomes. In the benchmark unit-demand case, this choice fully determines demand because each consumer purchases a fixed quantity. Many spatially competitive markets, however, also feature an important intensive margin: after choosing a supplier, consumers choose how much to buy. This paper develops tools for extending unit-demand spatial competition models to environments with general intensive-margin demand. The central tool is a reduction that summarizes the intensive-demand system by an indirect-surplus index and converts the spatial tariff problem into a unit-demand spatial game. The reduction is stated for a general spatial environment and does not depend on a particular geometry or equilibrium theorem.

The model below formalizes this idea in a two-stage spatial game. In stage one, heterogeneous firms choose locations and in stage two they choose two-part tariffs, each consisting of a marginal price and a fixed fee. Consumers are indexed by both a physical location and a characteristic affecting their fixed match terms with each supplier. These terms are

measured in utility-cost units and enter utility additively: higher values are worse for the consumer, while lower or negative values can represent better match quality or amenities. They may depend on distance-related costs that may vary across consumers in the same location because of consumer-specific weights on different dimensions of space, firm-specific amenities, or consumer-firm match components. The distribution of consumers across locations is arbitrary, as is the distribution of their characteristics at a given location. Consumers choose both which firm to buy from and, conditional on that choice, how much to buy. The reduction relies on two assumptions. First, consumer-firm fixed match terms may vary flexibly with consumer characteristics, firm identities, and firm locations, but they do not affect the delivered marginal price. Consumers served by the same seller therefore face the same marginal price and make the same intensive-margin quantity choice. Second, sellers can use two-part tariffs. The fixed fee allows a firm to change the marginal price while offsetting the induced change in consumer surplus, leaving unchanged the way the firm's tariff enters consumers' supplier-choice problem. Together, these assumptions allow the intensive-demand system to be summarized by a single surplus index.

These assumptions are natural when consumer-firm match quality terms reflect one-time search, travel, installation, contracting, compatibility, relationship-specific costs, or fixed match amenities, and when firms can charge fixed fees or rebates in addition to marginal prices. They are less appropriate when spatial frictions primarily affect the delivered marginal price of each unit consumed, as with transport costs that scale directly with quantity, weight, or volume.

Given these assumptions, the key tariff object is the tariff index. Let $v(p)$ denote the consumer's gross indirect surplus from buying from a firm charging marginal price p , before fixed fees and consumer-firm fixed match terms. A firm that charges marginal price p_i and fixed fee F_i enters consumers' spatial choice problem through its tariff index

$$z_i = F_i - v(p_i).$$

The spatial allocation of consumers to suppliers is determined by each consumer choosing the supplier with the lowest sum of the supplier-specific tariff index and the consumer-firm fixed match term. The scalar z_i therefore summarizes the firm-level tariff component of spatial attractiveness, while the marginal price p_i governs how much each captured consumer buys.

The reduction theorem separates a local implementation step from a game-level implication. Fix z_i . Among all two-part tariffs that implement this same tariff index, firm i maximizes per-consumer profit by choosing the *marginal-cost implementation* of z_i : it sets the marginal price equal to marginal cost and adjusts the fixed fee to keep the tariff index unchanged. This replacement leaves every consumer's net surplus from the firm unchanged, and therefore leaves all extensive-margin spatial choices unchanged. It also eliminates the distortion in the consumer's quantity choice and maximizes the transaction surplus generated by each captured consumer. This local two-part-tariff logic is familiar. The contribution is to show that, with additive fixed consumer-firm match terms, this point-wise implementation result applies after every location profile, turning a local two-part-tariff argument into a reduction of the spatial pricing game.

This implication is what makes the spatial problem tractable. Every tariff choice induces some tariff index, and the marginal-cost implementation of that same index weakly improves the firm’s payoff. Therefore no profitable deviation is lost by replacing the full tariff space with the one-dimensional space of tariff indices. Under the marginal-cost implementation, firm i ’s per-consumer margin takes the affine form $z_i + v(c_i)$, exactly as in a unit-demand spatial game after reinterpreting a high-surplus firm as a low-cost firm. General intensive demand enters the spatial game only through the surplus index $v(c_i)$, the surplus generated when firm i prices at marginal cost. Thus the paper’s reduction is not simply the marginal-cost pricing logic of two-part tariffs; it is the game-level reduction from the full tariff space to tariff indices, which makes the spatial pricing game with general intensive demand equivalent to a unit-demand spatial game with transformed firm types. Because this equivalence holds after every possible location profile, and because the location stage is unchanged, the equivalence extends from the pricing subgames to the full two-stage location-and-pricing game.

The formal result is stated in a general spatial environment. Each consumer is indexed by physical location and a characteristic affecting his fixed match terms with each supplier, each of which is distributed arbitrarily; firms may choose locations from an arbitrary feasible location set; and consumer-firm fixed match terms may be any measurable function of the consumer type, firm identity, and firm location. No common distance metric is needed for the reduction.

The reduction does *not* itself provide an equilibrium-existence theorem for every spatial environment. Its role is conditional: once the associated unit-demand spatial game has been solved, the reduction transfers that equilibrium characterization back to the two-part-tariff environment with general intensive demand.

To demonstrate the usefulness of the general results, we provide as a specific application a new spatial game. Heterogeneous firms choose their locations on the circumference of a circle in stage one and choose two-part tariffs in stage two. Consumers are uniformly distributed on the circumference of the circle and have arbitrary intensive-margin demand. Applying the reduction yields an associated unit-demand game: the circular model studied by [Vogel \(2008\)](#). Using our general results, we map the subgame-perfect equilibria characterized in [Vogel \(2008\)](#) back to obtain an equilibrium characterization of the two-part-tariff, general intensive-margin demand environment.

We contribute to the literature on nonlinear pricing in differentiated markets. The classic two-part tariff logic goes back to [Oi \(1971\)](#): marginal prices can govern quantities, while fixed fees transfer surplus. We use this standard implementation logic as an input; the contribution is the game-level reduction from tariffs to tariff indices in a spatial setting. In markets with differentiated firms and strategic pricing, [Armstrong and Vickers \(2001\)](#), [Rochet and Stole \(2002\)](#), [Armstrong and Vickers \(2010\)](#), and [Stole \(2007\)](#) study how nonlinear pricing and price discrimination more generally interact with product differentiation. The reduction from two-part tariffs to tariff indices is related to the utility-space approach of [Armstrong and Vickers \(2001\)](#). They show that, in differentiated-product settings, firms’ nonlinear offers can be summarized by the utility delivered to consumers, and that equilibrium outcomes

can be studied in this utility space. We use the same broad insight: in the present model, a two-part tariff induces a scalar tariff index. The primary contribution is to show that, with additive fixed consumer-firm match terms, this scalar index enters consumer choice exactly like a unit-demand price. Hence, after a change of variables, the reduced tariff-index game is payoff-equivalent to a single-price unit-demand spatial game with transformed firm types. Because this equivalence holds after every location profile, it applies to the full two-stage location-and-pricing game.

There is a substantial literature combining nonlinear pricing (and price discrimination more generally) with spatial differentiation, including [Spulber \(1981\)](#), [Stole \(1995\)](#), [Coyte and Lindsey \(1988\)](#), [Hamilton and Thisse \(1992, 1996\)](#), [Yin \(2004\)](#), [Hoernig and Valletti \(2007\)](#), and [Vogel \(2011\)](#). Relative to this literature, our contribution is not the marginal-cost pricing logic of two-part tariffs itself. The contribution is to show that, with additive fixed consumer-firm match terms and common two-part tariffs, general intensive demand can be absorbed into an indirect-surplus index, so every tariff deviation can be represented as a tariff-index deviation. The resulting tariff-index game is not merely lower-dimensional: after a change of variables, it is payoff-equivalent to a unit-demand spatial game with transformed firm types.

We contribute to the literature on spatial competition by showing how unit-demand spatial models can be extended to allow variable quantities. The classic models of [Hotelling \(1929\)](#) and [Salop \(1979\)](#) study differentiated competition through location with unit demand. Our result is a reduction: when consumer-firm fixed terms enter additively and firms use common two-part tariffs, general intensive demand enters the spatial game only through an indirect-surplus index, and the reduced game is a unit-demand spatial game with transformed firm types. The reduction does not solve every location game anew; rather, it provides a way to transport available unit-demand equilibrium characterizations back to environments with intensive-margin demand and two-part tariffs. The application below illustrates this by transporting the heterogeneous-firm unit-demand equilibrium of [Vogel \(2008\)](#) to an environment with arbitrary intensive-margin demand and two-part tariffs.¹

Finally, the reduction used here is related in spirit to recent work in spatial economics that makes high-dimensional location problems tractable by identifying lower-dimensional objects. [Oberfield et al. \(2024\)](#) study the spatial expansion of multi-establishment firms. There, a discrete location problem is disciplined by geometric limiting arguments that summarize local profitability and spatial assignment. Our reduction is different. Instead of taking a spatial limit, additive fixed consumer-firm match terms and common tariffs make each firm’s intensive-margin choice common across the consumers it serves, even when consumers differ in consumer-firm fixed terms or in how they measure distance. Two-part tariffs then allow the resulting intensive-demand system to enter the spatial problem through a single surplus index. This suggests a complementary route for spatial models with rich intensive

¹[Barro \(2024\)](#) also introduces intensive-margin demand into a circular Hotelling model, studying symmetric firms with constant-elasticity demand and delivered per-unit transport costs in a small-transport-cost limit.

demand: when consumer-firm fixed terms enter additively and nonlinear tariffs are available, intensive demand can be absorbed into an indirect-surplus index, leaving a unit-demand spatial allocation problem to be solved with standard geometric tools.

2 Environment

Let $\mathcal{X}$ be a set of locations, interpreted as geographic locations or positions in product-characteristic space, and let $\mathcal{Y}$ be a set of consumer characteristics. Consumers are indexed by $\omega = (x, y) \in \Omega \subseteq \mathcal{X} \times \mathcal{Y}$, where $x \in \mathcal{X}$ is the consumer's location and $y \in \mathcal{Y}$ is a characteristic that may affect consumer-firm fixed match terms. Consumers are distributed on the measurable consumer space Ω according to a finite measure μ , with $0 < \mu(\Omega) < \infty$. The reduction uses the measurable structure of this space, not a common metric on $\mathcal{X}$. There are $n \geq 2$ firms indexed by $i = 1, \dots, n$, each with constant marginal cost $c_i > 0$.

Firms play a two-stage game. In stage one, firms simultaneously choose physical locations, ℓ_i , from a nonempty feasible set $\mathcal{L} \subseteq \mathcal{X}$. In stage two, after observing all firms' locations, $\ell \equiv (\ell_1, \dots, \ell_n)$, firms simultaneously choose two-part tariffs (p_i, F_i) with $p_i > 0$ and $F_i \in \mathbb{R}$, where p_i is a per-unit or marginal price and F_i is a fixed fee. A consumer $\omega = (x, y)$ who buys quantity q from firm i at location $\ell_i \in \mathcal{L}$ obtains utility

$$U(q) - p_i q - F_i - A_i(\omega, \ell_i). \quad (1)$$

We describe the two remaining components of equation (1), $A_i(\omega, \ell_i)$ and $U(q)$, in what follows.

For each firm i , $A_i(\omega, \ell_i) = A_i((x, y), \ell_i)$ is a measurable consumer-firm fixed match term, measured in utility-cost units. It is subtracted from utility, so higher values of A_i are worse for the consumer, while lower or negative values can represent fixed amenities or better match quality. The term may depend on the consumer's location, the firm's identity, the firm's location, and the consumer characteristic y . It can include real fixed costs, such as travel, search, installation, contracting, compatibility, delivery setup, or other relationship-specific costs. It can also include firm amenities or consumer-firm match-quality components. Thus consumers at the same physical location may differ in how they evaluate the spatial or non-spatial fit of different firms. The reduction does not require a common metric over physical space. It requires only that each $A_i(\omega, \ell_i)$ enters utility additively and does not depend on quantity purchased. In such settings, the consumer faces a fixed consumer-firm match term when buying from firm i , while the marginal payment for additional units is the mill (or f.o.b.) price p_i .

The utility function $U(q)$ is defined over quantity and implicitly shapes intensive-margin demand, conditional on purchasing from a given firm. We allow for general intensive demand. Rather than imposing a parametric form for U , we impose the following conditions directly on the induced intensive demand function.²

²Conditional on having chosen a firm, a consumer solves $\max_{k \geq 0} \{U(k) - pk\}$. This yields $v(p) = \max_{k \geq 0} \{U(k) - pk\}$ and $q(p) \in \arg \max_{k \geq 0} \{U(k) - pk\}$.

Assumption 1. Conditional on choosing a firm that charges marginal price p , a consumer demands quantity $q(p) > 0$. The function $q(p)$ is weakly decreasing and locally integrable. Let v denote the associated indirect surplus function, normalized arbitrarily, satisfying

$$v(p') - v(p) = - \int_p^{p'} q(s) ds \quad (2)$$

for all $p, p' > 0$. Equivalently, where v is differentiable, $q(p) = -v'(p)$.

Consumers have an outside option that yields utility $\bar{u} \in \mathbb{R}$, where $\bar{u}$ is measured relative to the chosen normalization of v . A consumer purchases from a firm only if the best net surplus offered by firms is at least $\bar{u}$. Thus, under Assumption 1, a consumer ω choosing among firms compares $v(p_i) - F_i - A_i(\omega, \ell_i)$ across firms and also compares the best of these values with $\bar{u}$.

Equivalently, define firm i 's tariff index by

$$z_i \equiv F_i - v(p_i). \quad (3)$$

Then consumer ω chooses a firm with the lowest value of $z_i + A_i(\omega, \ell_i)$, provided that this lowest value is no greater than $-\bar{u}$. If

$$\min_j \{z_j + A_j(\omega, \ell_j)\} > -\bar{u},$$

the consumer chooses the outside option. Thus, for consumers' extensive-margin firm-choice decisions, the tariff index z_i is a sufficient statistic for firm i 's two-part tariff: all tariffs with the same z_i generate the same consumer assignment.

Ties are resolved by a fixed measurable tie-breaking rule, not specified here. For any location profile ℓ and any profile of tariff indices $z \equiv (z_1, \dots, z_n)$, define

$$a^*(\omega; \ell, z) \equiv \min_j \{z_j + A_j(\omega, \ell_j)\} \quad (4)$$

as the consumer-specific minimum value of $z_j + A_j(\omega, \ell_j)$ across all firms, where this value may be achieved from more than one supplier. The set of consumers supplied by firm i given locations and tariff indices is denoted by

$$B_i(\ell, z) = \left\{ \omega \in \Omega \left| \begin{array}{l} a^*(\omega; \ell, z) \leq -\bar{u}, \\ i \text{ is selected by the tie-breaking rule among} \\ \text{firms attaining } a^*(\omega; \ell, z) \end{array} \right. \right\}. \quad (5)$$

Since consumers are distributed according to the measure μ , the mass of consumers firm i sells to is

$$M_i(\ell, z) = \mu(B_i(\ell, z)). \quad (6)$$

An outcome is covered if $a^*(\omega; \ell, z) \leq -\bar{u}$ for μ -almost every ω . The reduction results below do not require coverage.³

³Applications that import a covered-market unit-demand equilibrium should impose conditions ensuring that the mapped equilibrium is covered.

Because consumer-firm match terms are fixed rather than per-unit costs, all consumers served by firm i face the same marginal price p_i . Conditional on choosing firm i , each consumer purchases $q(p_i)$ units, regardless of physical location x or characteristic y . Fixed consumer-firm match terms affect which consumers choose firm i , but not how much each captured consumer buys. Hence firm i 's profit is

$$\Pi_i = [F_i + (p_i - c_i)q(p_i)]M_i(\ell, z). \quad (7)$$

3 Reduction and Unit-Demand Representation

The proof proceeds in three steps. First, Lemma 1 establishes the local implementation result: holding fixed a firm's tariff index, setting marginal price equal to marginal cost maximizes per-consumer profit among all two-part tariffs that induce that index. Second, Proposition 1 uses this result to reduce the second-stage two-part-tariff problem to a single choice of a tariff index. Third, after a change of variables, Proposition 2 shows that the reduced tariff-index game is payoff-isomorphic to an associated unit-demand spatial game with transformed firm types.

3.1 Marginal-cost implementation

In the following lemma, we show that any tariff (p_i, F_i) can yield firm i no higher profits than an alternative tariff that keeps the same tariff index and sets the marginal price equal to c_i . In particular, the alternative tariff yields weakly higher profits per consumer and yields the same set of consumers, since the tariff indices in the original and alternative tariffs are equalized.

Lemma 1. *Fix a firm i and a tariff index z . For any marginal price $r > 0$, the unique fixed fee that implements the tariff index z is $F_z(r) = z + v(r)$. The per-consumer profit generated by the implementing tariff $(r, F_z(r))$ satisfies*

$$F_z(r) + (r - c_i)q(r) \leq z + v(c_i). \quad (8)$$

Equality is attained at $r = c_i$, in which case the implementing fixed fee is $F_z(c_i) = z + v(c_i)$. If q is strictly decreasing, equality is attained only at $r = c_i$.

Proof. For any marginal price r that implements the tariff index z , the fixed fee must be $F_z(r) = z + v(r)$. Hence per-consumer profit is

$$F_z(r) + (r - c_i)q(r) = z + v(r) + (r - c_i)q(r).$$

Substituting the previous equality into inequality (8), we obtain the inequality

$$v(r) + (r - c_i)q(r) \leq v(c_i). \quad (9)$$

We now prove inequality (9) is satisfied. If $r > c_i$, then by (2),

$$\begin{aligned} v(r) + (r - c_i)q(r) - v(c_i) &= - \int_{c_i}^r q(s) ds + (r - c_i)q(r) \\ &= - \int_{c_i}^r [q(s) - q(r)] ds \leq 0, \end{aligned}$$

because $q(s) \geq q(r)$ for $s \in [c_i, r]$. If $r < c_i$, then

$$\begin{aligned} v(r) + (r - c_i)q(r) - v(c_i) &= \int_r^{c_i} q(s) ds - (c_i - r)q(r) \\ &= \int_r^{c_i} [q(s) - q(r)] ds \leq 0, \end{aligned}$$

because $q(s) \leq q(r)$ for $s \in [r, c_i]$. At $r = c_i$ the expression equals zero. If q is strictly decreasing, then the relevant integral is strictly negative whenever $r \neq c_i$. $\square$

We can restate Lemma 1 more simply as follows: for any two-part tariff, firm i can weakly increase profit by setting the per-unit price equal to marginal cost and adjusting the fixed fee to leave consumers' net utility unchanged. The intuition behind Lemma 1 is straightforward. The per-unit price p affects how much a consumer buys. If $p \neq c_i$, the consumer's quantity choice is distorted away from the efficient level for the bilateral transaction between firm i and the consumer. The fixed fee F , by contrast, transfers surplus without distorting the quantity purchased. Thus, holding fixed the consumer's net surplus from buying from firm i , the firm does best by setting $p = c_i$, which maximizes the transaction surplus generated by that match, and then adjusting the fixed fee to leave the consumer's net utility unchanged. The inequality in Lemma 1 is exactly this transaction-surplus argument: $v(p) + (p - c_i)q(p) \leq v(c_i)$. The left side is the surplus generated by a consumer-firm i match when the firm charges marginal price p : consumer indirect surplus before the fixed fee, $v(p)$, plus firm variable profit, $(p - c_i)q(p)$. The right side is the same object when marginal price equals marginal cost, $p = c_i$. Fixed consumer-firm match terms and the consumer's alternatives do not enter this comparison because the replacement tariff leaves the consumer's net surplus from firm i unchanged, and hence leaves firm i 's attractiveness relative to all alternatives unchanged. Because q is weakly decreasing, marginal-cost pricing maximizes this transaction surplus.

For firm i and tariff index z_i , define the *marginal-cost implementation* of z_i by

$$\mathcal{C}_i(z_i) \equiv (c_i, z_i + v(c_i)). \tag{10}$$

The tariff in equation (10) implements the tariff index z_i , since $(z_i + v(c_i)) - v(c_i) = z_i$. By Lemma 1, among all two-part tariffs that implement z_i , the marginal-cost implementation maximizes per-consumer profit.

3.2 Reduction to a tariff-index game

We now define the reduced tariff-index game. It has the same first stage as the original game: firms choose locations $\ell_i \in \mathcal{L}$. In the second stage, after observing locations, firm i chooses a tariff index $z_i \in \mathbb{R}$. Consumers choose firms, or the outside option, according to equations (4) and (5). The reduced payoff of firm i is

$$\Pi_i^R(\ell, z) = (z_i + v(c_i))M_i(\ell, z). \quad (11)$$

This payoff is the profit firm i obtains in the original two-part-tariff game when the index z_i is implemented by the marginal-cost implementation $\mathcal{C}_i(z_i)$. In the reduced tariff-index game, however, firms choose only the index z_i , not a particular two-part-tariff implementation of that index.

The following proposition turns Lemma 1 into a reduction of the entire two-part tariff game. Lemma 1 is a point-wise implementation result: holding fixed a given tariff index, it identifies the profit-maximizing two-part tariff that implements that index. Proposition 1 shows that this fixed-index implementation result is enough to replace the full two-part-tariff problem by a one-dimensional tariff-index problem. For any location profile ℓ , every tariff deviation (p_i, F_i) induces some tariff index $\tilde{z}_i = F_i - v(p_i)$, and the marginal-cost implementation $\mathcal{C}_i(\tilde{z}_i)$ captures the same consumers while yielding weakly higher per-consumer profit. Hence checking profitable deviations in the original two-part-tariff game is equivalent to checking profitable deviations in the reduced game.

Proposition 1. *For fixed locations ℓ and rivals' tariff indices, the set of consumers captured by firm i depends on firm i 's tariff only through the tariff index $z_i = F_i - v(p_i)$. Among all two-part tariffs that induce a given tariff index z_i , firm i maximizes total profit by choosing the marginal-cost implementation*

$$(p_i, F_i) = \mathcal{C}_i(z_i) = (c_i, z_i + v(c_i)).$$

The resulting profit per captured consumer is $z_i + v(c_i)$, and the resulting total profit is the reduced payoff in (11). Consequently, for each fixed location profile ℓ , the marginal-cost implementation profile

$$(\mathcal{C}_i(z_i))_{i=1}^n = ((c_i, z_i + v(c_i)))_{i=1}^n$$

is a Nash equilibrium of the original two-part-tariff pricing subgame if and only if $z = (z_1, \dots, z_n)$ is a Nash equilibrium of the reduced tariff-index game with payoffs (11).

Proof. Consumer choice depends on a tariff (p_i, F_i) only through the tariff index $z_i = F_i - v(p_i)$, as in (3). Hence, holding locations and rivals' tariff indices fixed, all tariffs that induce the same z_i capture the same set of consumers.

Fix such a tariff index z_i . Lemma 1 implies that the largest per-consumer profit attainable among tariffs inducing z_i is $z_i + v(c_i)$, attained by setting $p_i = c_i$ and $F_i = z_i + v(c_i)$. Since market mass depends only on locations and the vector of tariff indices z , total profit is given by (11).

It remains to justify that the reduction does not omit profitable tariff deviations. Consider any two-part-tariff deviation (p_i, F_i) in the original game, holding fixed the rivals' tariffs and hence their tariff indices z_{-i} . Let $\tilde{z}_i = F_i - v(p_i)$ be the tariff index induced by this deviation. The deviating tariff (p_i, F_i) and the marginal-cost implementation $\mathcal{C}_i(\tilde{z}_i)$ induce the same tariff index $\tilde{z}_i$. Hence, given rivals' tariff indices z_{-i} and the location profile ℓ , they capture the same consumers, namely the consumers in firm i 's market under the tariff-index profile $(\tilde{z}_i, z_{-i})$. The deviating tariff's per-consumer profit is

$$F_i + (p_i - c_i)q(p_i) = \tilde{z}_i + v(p_i) + (p_i - c_i)q(p_i) \leq \tilde{z}_i + v(c_i),$$

where the inequality follows from Lemma 1. Thus firm i 's payoff from the tariff deviation is no greater than $(\tilde{z}_i + v(c_i))M_i(\ell, \tilde{z}_i, z_{-i})$. This bound is exactly the payoff from the tariff-index deviation $\tilde{z}_i$ in the reduced game, attained in the original tariff game by choosing $\mathcal{C}_i(\tilde{z}_i)$. Hence, if some two-part-tariff deviation is profitable in the original game, then the corresponding tariff-index deviation is profitable in the reduced game.

Conversely, if a tariff-index deviation to $\tilde{z}_i$ is profitable in the reduced game, then firm i can implement it in the original game by choosing the marginal-cost implementation $\mathcal{C}_i(\tilde{z}_i)$. Hence checking profitable tariff-index deviations is equivalent to checking profitable two-part-tariff deviations.

The argument also applies to mixed strategies. To see this, consider any mixed tariff strategy for firm i . Each tariff (p_i, F_i) in the support of this mixed strategy induces a tariff index $z_i = F_i - v(p_i)$. Replace each such tariff by the marginal-cost implementation $\mathcal{C}_i(z_i) = (c_i, z_i + v(c_i))$ that induces the same tariff index. This replacement leaves the distribution of tariff indices unchanged, so it leaves the distribution of consumer assignments unchanged. By Lemma 1, it weakly raises firm i 's profit for every realized tariff in the support. Therefore it also weakly raises firm i 's expected profit. Hence the reduction preserves best-response inequalities for mixed strategies as well as pure strategies. $\square$

Discussion. Proposition 1 is the step that turns the standard two-part-tariff implementation logic of Oi (1971) into a spatial-game reduction. Lemma 1 is a local statement: holding fixed a tariff index, the marginal-cost implementation maximizes per-consumer profit. Proposition 1 is stronger: it implies that no profitable deviation is lost by replacing the full tariff space with the one-dimensional space of tariff indices. The next subsection introduces the associated unit-demand spatial game, and Proposition 2 shows that the reduced tariff-index game is payoff-isomorphic to it after a change of variables.

Two components of the reduction are easy to conflate. Additive fixed consumer-firm match terms and common firm-level tariffs imply that all consumers served by a given firm face the same marginal price. Hence they buy the same quantity and generate the same per-consumer operating margin, regardless of physical location or fixed-cost characteristic. This part of the argument does not require the marginal-cost implementation; it follows from additive fixed consumer-firm match terms and common tariffs. It does, however, use the assumption that the intensive-demand system is common across consumers. Nonuniform consumer density and

heterogeneous fixed consumer-firm match terms, including type-specific notions of distance and fixed match components, are allowed because they change the measure and composition of the set served by a firm, not the quantity purchased by each consumer served by that firm. By contrast, if different consumer types had different indirect-surplus functions and chose suppliers individually (i.e., there was no local aggregation into a representative consumer), a single fixed fee could not generally offset a change in marginal price for all types at once. Here, “different indirect-surplus functions” means differences in the p -dependent part of indirect surplus, equivalently differences in intensive demand. It does not refer merely to additive type-specific or consumer-firm-specific terms that are independent of quantity; those terms are already allowed in the fixed consumer-firm match term $A_i(\omega, \ell_i)$. The assumption that fixed consumer-firm match terms enter additively is also doing important work. If the spatial cost component of the consumer-firm term instead entered the marginal price, consumer surplus from firm i would take the form $v(p_i + \tilde{A}_i(\omega, \ell_i)) - F_i$. For this expression to have the additive-index form needed for the reduction, it would have to be written as a tariff-specific scalar plus a consumer-firm match term that may vary with ω and ℓ_i but is independent of p_i . Equivalently, the effect of changing p_i on consumer surplus would have to be the same for all consumers served by the firm. Where v is differentiable, this effect is governed by

$$\frac{\partial}{\partial p_i} v(p_i + \tilde{A}_i(\omega, \ell_i)) = -q(p_i + \tilde{A}_i(\omega, \ell_i)),$$

which generally varies with ω and ℓ_i . It is independent of the consumer-firm match term only in the unit-demand case, or more generally when q is constant over the relevant range. Thus, except when intensive demand is constant over the relevant range, spatial costs that enter the delivered marginal price destroy the additive structure needed for the tariff-index reduction.

The role of the two-part tariff is different: it lets the firm change the marginal price without changing its spatial attractiveness, because the fixed fee can offset the resulting change in indirect surplus. The marginal-cost implementation uses this freedom to set the marginal price equal to marginal cost while preserving the same tariff index.

Proposition 1 completes the first step of the argument. It replaces the full two-part-tariff pricing problem with a second-stage game in which each firm chooses only a scalar tariff index z_i and earns payoff given by (11). The spatial environment is otherwise unchanged: locations, consumer measure, fixed consumer-firm match terms, tie-breaking, and outside-option choices are all inherited from the original model. The next subsection introduces the associated unit-demand spatial game and the subsequent subsection shows that this reduced tariff-index game is exactly payoff-isomorphic to it after a change of variables.

3.3 The associated unit-demand game

Proposition 1 converts the original two-part-tariff game into a reduced game in which firms choose tariff indices in the second stage. We now introduce a distinct two-stage game on the same spatial environment: an associated unit-demand price game. This game is not an additional model assumption. Rather, Proposition 2 below shows that, after a change of

variables, this associated unit-demand game is payoff-isomorphic to the reduced tariff-index game.

Fix a constant $K \in \mathbb{R}$. The associated unit-demand game has the same consumer space Ω , consumer measure μ , feasible location set $\mathcal{L}$, fixed consumer-firm match term functions $(A_i)_{i=1}^n$, and tie-breaking rule as the original two-part-tariff game. It differs in the pricing and demand primitives. Consumers have unit demand, firms each choose a single price rather than a two-part tariff, and firms have transformed marginal costs. Firms again choose locations in stage one. In stage two, after observing locations, firm i chooses a price $\hat{p}_i \in \mathbb{R}$.

A consumer ω compares delivered prices $\hat{p}_i + A_i(\omega, \ell_i)$. The consumer buys one unit from a firm with the lowest delivered price provided that this delivered price is no greater than the associated reservation value

$$\hat{R} \equiv K - \bar{u}. \quad (12)$$

Otherwise, the consumer chooses the outside option. Let $\hat{M}_i(\ell, \hat{p})$ be the mass of consumers who choose firm i in the associated unit-demand game. Firm i 's transformed marginal cost is

$$\hat{k}_i \equiv K - v(c_i). \quad (13)$$

Its payoff is

$$\hat{\Pi}_i(\ell, \hat{p}) = (\hat{p}_i - \hat{k}_i) \hat{M}_i(\ell, \hat{p}). \quad (14)$$

The role of K is only to choose the price and cost levels in the associated unit-demand game. In the isomorphism below, a tariff index z_i will correspond to the unit-demand price $\hat{p}_i = z_i + K$. Together with (13), this gives

$$\hat{p}_i - \hat{k}_i = (z_i + K) - (K - v(c_i)) = z_i + v(c_i),$$

so unit-demand markups equal reduced-game per-consumer margins. The same shift also changes the outside-option cutoff from $-\bar{u}$ in the tariff-index formulation to $\hat{R} = K - \bar{u}$ in the associated unit-demand game.

3.4 Payoff isomorphism

We now compare the reduced tariff-index game from Proposition 1 to the associated unit-demand price game. Given the constant K fixed above, define the price transformation

$$\hat{p}_i = z_i + K. \quad (15)$$

Together with the transformed costs in (13), this transformation maps tariff-index payoffs into unit-demand price payoffs.

Proposition 2. *Under the change of variables in equation (15) and the transformed costs in (13), the reduced tariff-index two-stage game is payoff-isomorphic to the associated unit-demand price two-stage game. Consumer assignments, including choices of the outside option, coincide for every location profile and every second-stage strategy profile. Each firm's payoff*

is identical. Consequently, Nash equilibria of corresponding second-stage continuation games, and subgame-perfect equilibria of the two-stage games, are preserved under the change of variables.

Proof. Fix any location profile ℓ . For any consumer ω and firm i , the delivered price in the associated unit-demand game is

$$\widehat{p}_i + A_i(\omega, \ell_i) = z_i + K + A_i(\omega, \ell_i).$$

The added constant K is common across firms. Hence the firm minimizing $\widehat{p}_i + A_i(\omega, \ell_i)$ is exactly the firm minimizing $z_i + A_i(\omega, \ell_i)$.

The outside-option decisions also coincide. Since

$$\min_j \{\widehat{p}_j + A_j(\omega, \ell_j)\} = K + \min_j \{z_j + A_j(\omega, \ell_j)\},$$

we have

$$\min_j \{\widehat{p}_j + A_j(\omega, \ell_j)\} \leq K - \bar{u} \iff \min_j \{z_j + A_j(\omega, \ell_j)\} \leq -\bar{u}.$$

Thus consumers make the same purchase-or-outside-option decision in the two games and, conditional on purchasing, choose the same firm under the common tie-breaking rule. Therefore

$$\widehat{M}_i(\ell, \widehat{p}) = M_i(\ell, z).$$

Payoffs also coincide because

$$\widehat{p}_i - \widehat{k}_i = (z_i + K) - (K - v(c_i)) = z_i + v(c_i).$$

Hence

$$\widehat{\Pi}_i(\ell, \widehat{p}) = (\widehat{p}_i - \widehat{k}_i) \widehat{M}_i(\ell, \widehat{p}) = (z_i + v(c_i)) M_i(\ell, z) = \Pi_i^R(\ell, z).$$

The transformation in equation (15) gives a one-to-one correspondence between tariff-index profiles $z \in \mathbb{R}^n$ and unit-demand price profiles $\widehat{p} \in \mathbb{R}^n$. Because consumer assignments, outside-option choices, and payoffs coincide under this correspondence after every location profile, best-response inequalities and Nash equilibria are preserved in every second-stage continuation game. The same conclusion holds for mixed strategies, since the correspondence can be applied to each realized tariff-index profile and preserves expected payoffs. Since the first-stage location sets are also identical, subgame-perfect equilibria of the reduced tariff-index game and the associated unit-demand game correspond, including equilibria with mixed continuation strategies. $\square$

Combining Proposition 1 with Proposition 2 gives the comparison between the original two-part-tariff game and the associated unit-demand price game. Proposition 1 shows that, after every location profile, the original tariff subgame can be reduced without loss to the tariff-index subgame. Proposition 2 shows that this reduced subgame is payoff-isomorphic to the associated unit-demand price subgame. Since both results hold after every possible

first-stage location profile, they preserve not only on-path pricing incentives but also off-path continuation payoffs after location deviations.

Thus any subgame-perfect equilibrium of the associated unit-demand price game maps into a subgame-perfect equilibrium of the original two-part-tariff game by setting

$$z_i = \hat{p}_i - K$$

and then using the marginal-cost implementation

$$(p_i, F_i) = \mathcal{C}_i(z_i) = (c_i, z_i + v(c_i)).$$

Conversely, any subgame-perfect equilibrium of the original game whose second-stage tariffs are marginal-cost implementations induces a subgame-perfect equilibrium of the associated unit-demand price game through $\hat{p}_i = z_i + K$.

If a target unit-demand model restricts prices to be nonnegative, choose $K > \max_i v(c_i)$ and restrict attention in the reduced tariff-index game to $z_i \geq -K$. Then the transformation $\hat{p}_i = z_i + K$ maps the restricted tariff-index strategy space bijectively onto the nonnegative price space $\hat{p}_i \geq 0$. This restriction is without loss for equilibrium and profitable-deviation checks in the reduced game: if $z_i < -K$, then $z_i + v(c_i) < 0$, so firm i earns negative per-consumer profit whenever it serves a positive mass of consumers, while it can always choose a tariff index sufficiently high to serve no consumers and earn zero profit.

Discussion. Propositions 1 and 2 are transfer results. They do not solve the associated unit-demand location game. Instead, they show that any equilibrium characterization of the associated unit-demand game can be transported back to the original two-part-tariff game with general intensive demand.

4 Application to the Circular Model of Vogel (2008)

We now apply this transfer result to a new spatial environment for which the associated unit-demand game has already been solved. The application is the circular model of Vogel (2008), specialized to zero firm shipping cost. Once the unit-demand equilibrium characterization is imported, Propositions 2 and 1 map it back to the two-part-tariff environment.

In this application, the physical location space is the unit circle,

$$\mathbb{T} \equiv \mathbb{R}/\mathbb{Z}.$$

Let $d_{\mathbb{T}}(x, x')$ denote circular distance on $\mathbb{T}$. There is no heterogeneity in fixed-cost characteristics in this application, so consumers can be identified with their physical locations: $\Omega = \mathbb{T}$. Consumers are uniformly distributed on $\mathbb{T}$ with total mass $L > 0$, so the consumer measure is $\mu = L\lambda$, where λ is the Lebesgue measure on the unit circle normalized so that $\lambda(\mathbb{T}) = 1$. Firms may locate anywhere on the circle, so $\mathcal{L} = \mathbb{T}$. There are no firm-specific

fixed amenities in this application, and the consumer-firm fixed match terms are linear access costs in circular distance:

$$A_i(\omega, \ell_i) = t d_{\mathbb{T}}(\omega, \ell_i), \quad t > 0.$$

Thus the associated unit-demand game has the same unit circle, the same consumer mass L , and the same consumer-firm fixed match term $t d_{\mathbb{T}}(\omega, \ell_i)$ as the corresponding unit-demand model.

We import the following result from [Vogel \(2008\)](#), specialized to zero firm shipping cost.

Theorem 1 ([Vogel \(2008\)](#), specialized to zero firm shipping cost). *Consider the unit-demand spatial game on the unit circle with consumer transportation cost t , consumer mass L , marginal costs $(\widehat{k}_i)_{i=1}^n$, and zero firm shipping cost. For every $\kappa \geq 0$ there exists a number*

$$\varphi = \varphi(n, t, 0, L, \kappa) > 0$$

such that if $\widehat{k}_i \in [\kappa, \kappa + \varphi]$ for every firm i , then for any cyclic order of the $n \geq 2$ firms there exists a subgame-perfect equilibrium with pure strategies on the equilibrium path. In this equilibrium, adjacent distances, prices, consumer shares, and profits are

$$\widehat{D}_{i,i+1}^* = \frac{1}{n} + \frac{2}{3t} \left(\widehat{k} - \frac{\widehat{k}_i + \widehat{k}_{i+1}}{2} \right), \quad (16)$$

$$\widehat{p}_i^* = \frac{t}{n} + \frac{2}{3} \widehat{k} + \frac{1}{3} \widehat{k}_i, \quad (17)$$

$$\widehat{s}_i^* = \frac{1}{n} + \frac{2}{3t} \left(\widehat{k} - \widehat{k}_i \right), \quad (18)$$

$$\widehat{\Pi}_i^* = Lt(\widehat{s}_i^*)^2. \quad (19)$$

We now map this unit-demand equilibrium back to the two-part-tariff environment with arbitrary intensive demand. Fix $\kappa > 0$ and choose

$$K \equiv \kappa + \max_i v(c_i). \quad (20)$$

The associated unit-demand cost of firm i is

$$\widehat{k}_i = K - v(c_i).$$

Let

$$\bar{v}(c) \equiv \frac{1}{n} \sum_{i=1}^n v(c_i) \quad \text{and} \quad \rho \equiv \frac{3t}{2n} - \bar{v}(c), \quad (21)$$

where $\bar{v}(c)$ is the average indirect surplus across firms and the scalar ρ is the common value of the distance-adjusted tariff index at every boundary between adjacent firms' market segments in the equilibrium constructed below. Define firm i 's half-market length by

$$h_i \equiv \frac{\rho + v(c_i)}{3t}. \quad (22)$$

The following two assumptions state the relevant similarity condition and the covered-market condition.

Assumption 2. The transformed unit-demand costs satisfy the sufficient similarity condition:

$$K - v(c_i) \in [\kappa, \kappa + \varphi(n, t, 0, L, \kappa)] \quad \text{for all } i.$$

Equivalently,

$$\max_i v(c_i) - \min_i v(c_i) \leq \varphi(n, t, 0, L, \kappa).$$

Assumption 3. The outside option is low enough that

$$\bar{u} \leq -\rho.$$

Assumption 3 says that even boundary consumers, who receive the lowest net surplus from their chosen firm in the centered equilibrium, prefer buying from some firm to taking the outside option. Equivalently, in the associated unit-demand game, the reservation delivered price $\hat{R} = K - \bar{u}$ is at least the common boundary delivered price $K + \rho$.

At this point, the circular equilibrium characterization follows by translating the associated unit-demand equilibrium back to the two-part-tariff environment. Under Assumption 2, the transformed costs $\hat{k}_i = K - v(c_i)$ satisfy the hypotheses of Theorem 1. Hence, for any cyclic order, Theorem 1 gives a subgame-perfect equilibrium of the associated unit-demand price game. Proposition 2 maps that equilibrium into an equilibrium of the reduced tariff-index game, and Proposition 1 then maps the reduced equilibrium back into the original two-part-tariff game by using the marginal-cost implementation. The following corollary records the resulting equilibrium formulas. The proof is an algebraic translation of the formulas in Theorem 1 and is given in the Appendix.

Corollary 1. *Suppose Assumptions 1, 2, and 3 hold. For any cyclic order of the $n \geq 2$ firms on the unit circle, there is a subgame-perfect equilibrium of the two-part-tariff game with pure strategies on the equilibrium path.*

In this equilibrium, the distance between any two neighbors i and $i + 1$ is

$$D_{i,i+1}^* = \frac{1}{n} + \frac{v(c_i) + v(c_{i+1}) - 2\bar{v}(c)}{3t} = h_i + h_{i+1}, \quad (23)$$

where indices are modulo n . Firm i 's on-path two-part tariff is

$$(p_i^*, F_i^*) = \left(c_i, \frac{2(\rho + v(c_i))}{3} \right), \quad (24)$$

and its tariff index is

$$z_i^* = F_i^* - v(c_i) = \frac{2\rho - v(c_i)}{3}. \quad (25)$$

Firm i 's consumer share, consumer mass, total quantity, profit, and sales share are

$$s_i^{C,*} = 2h_i = \frac{2(\rho + v(c_i))}{3t}, \quad (26)$$

$$M_i^* = Ls_i^{C,*} = \frac{2L(\rho + v(c_i))}{3t}, \quad (27)$$

$$Q_i^* = q(c_i)M_i^*, \quad (28)$$

$$\Pi_i^* = F_i^* M_i^* = \frac{4L(\rho + v(c_i))^2}{9t}, \quad (29)$$

$$s_i^{Q,*} = \frac{q(c_i)M_i^*}{\sum_{k=1}^n q(c_k)M_k^*}. \quad (30)$$

The corollary inherits the order-invariance property of the associated unit-demand game. For a fixed set of firms, firm i 's marginal price, fixed fee, consumer share, consumer mass, quantity, sales share, and profit depend on firm i 's own primitives and on the multiset of firm primitives, but not on the cyclic order of firms or on the identity of firm i 's neighbors. Neighbor identities affect the adjacent distances in (23), but not the firm-level outcomes in (24)-(30).

5 Conclusion

This paper develops a general reduction that allows intensive-margin demand to be incorporated into spatial competition models while preserving the structure of unit-demand spatial games. The marginal-cost pricing logic of two-part tariffs is the local implementation step. The game-level implication is that, with additive fixed consumer-firm match terms and common two-part tariffs, no profitable deviation is lost by replacing the full tariff space with the one-dimensional space of tariff indices. General intensive demand is then summarized by the surplus index $v(c_i)$, and the reduced spatial pricing problem is payoff-equivalent to an associated unit-demand game with transformed firm types. Because the reduction holds after every location profile, the equivalence applies to the full location-and-pricing game, not only to the on-path pricing subgame.

The reduction is modular. It can be applied to any spatial environment in which consumer-firm fixed terms enter consumer choice additively and firms use common two-part tariffs. In such environments, each tariff induces a firm-level tariff index, and the intensive-demand system can be summarized by the surplus index $v(c_i)$. Applied to the circular model of Vogel (2008), the reduction yields a subgame-perfect equilibrium characterization with arbitrary intensive demand and two-part tariffs.

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APPENDIX

Proof of Corollary 1.

Proof. By Assumption 2, the transformed costs $\widehat{k}_i = K - v(c_i)$ satisfy the hypothesis of Theorem 1. Hence, for any cyclic order, the associated unit-demand price game has a subgame-perfect equilibrium with pure strategies on the equilibrium path. By Proposition 2, this equilibrium maps into a subgame-perfect equilibrium of the reduced tariff-index game. By Proposition 1, the reduced equilibrium maps into a subgame-perfect equilibrium of the original two-part-tariff game by using the marginal-cost implementation

$$(p_i^*, F_i^*) = \mathcal{C}_i(z_i^*) = (c_i, z_i^* + v(c_i)).$$

It remains only to translate the formulas. Since $\widehat{k}_i = K - v(c_i)$,

$$\widehat{k} = K - \bar{v}(c).$$

Using (16),

$$\begin{aligned} D_{i,i+1}^* &= \frac{1}{n} + \frac{2}{3t} \left(K - \bar{v}(c) - \frac{K - v(c_i) + K - v(c_{i+1})}{2} \right) \\ &= \frac{1}{n} + \frac{v(c_i) + v(c_{i+1}) - 2\bar{v}(c)}{3t}. \end{aligned}$$

Using (21) and (22),

$$h_i + h_{i+1} = \frac{2\rho + v(c_i) + v(c_{i+1})}{3t} = \frac{1}{n} + \frac{v(c_i) + v(c_{i+1}) - 2\bar{v}(c)}{3t},$$

so (23) follows. If $n = 2$, then $v(c_1) + v(c_2) = 2\bar{v}(c)$, so each of the two arcs has length $1/2$.

By (17),

$$\begin{aligned} z_i^* + K &= \widehat{p}_i^* = \frac{t}{n} + \frac{2}{3}(K - \bar{v}(c)) + \frac{1}{3}(K - v(c_i)) \\ &= K + \frac{t}{n} - \frac{2}{3}\bar{v}(c) - \frac{1}{3}v(c_i). \end{aligned}$$

Thus

$$z_i^* = \frac{t}{n} - \frac{2}{3}\bar{v}(c) - \frac{1}{3}v(c_i) = \frac{2\rho - v(c_i)}{3},$$

which is (25). The marginal-cost implementation then gives

$$p_i^* = c_i, \quad F_i^* = z_i^* + v(c_i) = \frac{2(\rho + v(c_i))}{3},$$

which is (24).

Using (18),

$$\begin{aligned} s_i^{C,*} &= \frac{1}{n} + \frac{2}{3t} (K - \bar{v}(c) - (K - v(c_i))) \\ &= \frac{1}{n} + \frac{2(v(c_i) - \bar{v}(c))}{3t} = \frac{2(\rho + v(c_i))}{3t} = 2h_i, \end{aligned}$$

which proves (26). Because consumers have total mass L , firm i 's consumer mass is

$$M_i^* = L s_i^{C,*},$$

which gives (27). Since $p_i^* = c_i$, each captured consumer buys $q(c_i)$, so total quantity is (28). Firm i earns only its fixed fee per captured consumer because the marginal price equals marginal cost. Hence

$$\Pi_i^* = F_i^* M_i^* = \frac{2(\rho + v(c_i))}{3} \cdot \frac{2L(\rho + v(c_i))}{3t} = \frac{4L(\rho + v(c_i))^2}{9t},$$

which is (29). The sales-share formula follows from

$$s_i^{Q,*} = \frac{Q_i^*}{\sum_{k=1}^n Q_k^*}.$$

Finally, Assumption 3 ensures that the constructed equilibrium is covered. At the boundary of firm i 's market, the distance-adjusted tariff index is

$$z_i^* + th_i = \frac{2\rho - v(c_i)}{3} + \frac{\rho + v(c_i)}{3} = \rho.$$

Thus every boundary consumer obtains net surplus $-\rho$. Consumers inside firm i 's market are closer to firm i and therefore obtain weakly higher net surplus. Since $\bar{u} \leq -\rho$, all consumers buy from some firm. $\square$